



TRAUMA CENTER STANDARDS

Effective: February 15, 2025
Replaces: March 10, 2024

I. Purpose

The provide the procedures and standards for the Santa Clara County Trauma Care System. This system links pre-hospital and hospital care to ensure patients with traumatic injuries are triaged and prioritized to receive care at the closest trauma center and receive optimal care based on these standards.

II. Authority

California Health and Safety Code, Division 2.5, Sections 1797.107, 1798, 1798.161, 1798.160, 1798.163, 1798.165

California Code of Regulations Title 22, Division 9, Chapter 1.1

California Code of Regulations Title 22, Division 9, Chapter 6.1

III. Definitions

- A. "American College of Surgeons Committee on Trauma" (ACS-COT) is an organization focused on improving care of injured patients through education, advocacy, trauma system development, and trauma center accreditation. They have published a trauma center standards manual *Resources for Optimal Care of the Injured Patient* (2023), which details criteria for trauma center verification.
- B. "Continuously available" means 24 hours a day, 7 days a week, 365 days a year (24/7/365) and implies there are no gaps in coverage.
- C. "Trauma Alert Patient" means a person, adult or pediatric deemed a trauma patient under the trauma triage criteria set forth in the EMS Agency's policies and procedures.
- D. "Trauma Center" means a licensed hospital which has been designated as a Level I or II trauma center and/or Level I or II pediatric trauma center by the local EMS agency.
 - 1. Level I (L1) are adult trauma centers capable of providing system leadership and comprehensive trauma care for all injuries. It must have adequate depth



of resources and personnel to provide optimal patient care, education, and conduct research. These centers have an important role in collaborating with the EMS Agency in trauma system development, regional disaster planning, increasing capacity, and advancing trauma care.

2. Level II (L2) trauma centers are expected to provide initial definitive trauma care for a wide range of injuries and injury severity and may take on additional responsibilities on the region related to education, system leadership, and disaster planning.
3. A pediatric trauma center (PTC) offers the same services as adult centers but is focused on providing care to patients less than 15 years of age and are designated as either Level I (PTC1) or Level II (PTC2).

IV. The Santa Clara County EMS Agency (EMS Agency)

A. The EMS Agency and EMS Medical Director are responsible for the oversight and management of the Trauma System including but not limited to planning, implementation and evaluation. Areas of responsibility include:

1. Trauma System design and planning.
2. Designation and monitoring for compliance with standards.
3. Medical control, clinical operations, training, and management of the trauma system in the pre-hospital care setting.
4. Developing pre-hospital trauma triage and destination policies.
5. Coordinating trauma system quality improvement.

V. Designation

A. The process for initial designation as a trauma center by Santa Clara County Emergency Medical Services are detailed in Policy #400.

B. Continued designation as trauma center in Santa Clara County shall depend on:

1. Current written agreement with Santa Clara EMS for designation as a Trauma Center.



2. Currently serving in the EMS System as a Pre-hospital Receiving Center (PRC).
3. Maintain verification from ACS-COT ensuring continuous compliance with most current L1, L2, PTC1 or PTC2 standards.
 - a. A copy of the renewal verification report shall be provided to the EMS Agency within thirty (30) days of receipt.
4. Compliance with all State regulations and Santa Clara County EMS Agency policies and procedures including those detailed herein.
5. If there is any conflict between the State regulations, local EMS policies and procedures, or ACS-COT verification standards, the more stringent rule will apply until next policy review.

VI. Trauma Center Standards

A designated trauma center shall meet all the following standards:

A. Hospital Facility and Services

1. Have continuously available for the evaluation and care of trauma alert patients, a trauma surgeon, neurosurgeon, orthopedic surgeon, anesthesia, critical care physician, emergency department physician, and trauma resuscitation nurse.
2. Maintain equipment for resuscitation and diagnosis to provide advanced life support for the critically and seriously injured adult and pediatric patient.
3. Hospital staff on the trauma team and in support services shall be certified and trained according to current ACS-COT verification standards.
4. Surgical, non-surgical specialist, nursing, and support service availability shall meet the current ACS-COT verification standards for the appropriate designation level.
5. Written policies and procedures for the activation, triage, and diagnosis of both EMS and self-transport trauma alert patients.
6. Clinical practice guidelines (CPGs), protocols, policies, and procedures must align with the ACS-COT verification standards.



7. A process in place to triage and treat simultaneously arriving trauma patients.
8. Maintain uninterrupted trauma capabilities which include advanced imaging, laboratory services, and endovascular or interventional radiological services that are continuously available.
9. Operating room (OR) must be available for trauma patients within 30 minutes of notification.
10. Not exceed 100 hours of bypass per quarter.
11. Have a dedicated patient decontamination area, proper equipment, and staff trained to manage patients exposed to hazardous materials.
12. Heliport approved by California Division of Aeronautics, Department of Transportation.
13. Establish inter-facility transfer agreements with referring and specialty hospitals.
14. Have the capability to provide both telephone and on-site consultations with physicians in the community and outlying areas

B. Support Services

Support Services must be available within the timeframes established by ACS-COT for designation level.

1. Radiology and Imaging Services
2. Clinical Laboratory Services
3. Blood Bank Services
4. Surgical Services
5. Post Anesthesia Recovery Room
6. Intensive Care Unit
7. Renal Replacement Therapy Services



8. Pharmacy Services
9. Respiratory Therapy
10. Organ Procurement Program
11. Injury Prevention Program
12. Rehabilitation Services

C. Program Personnel

1. Trauma Program Medical Director (TPMD)
Designate a board-certified General Surgeon with sufficient experience and expertise in managing trauma patients. They provide medical oversight of the trauma program while also ensuring compliance with all applicable statutes, regulations and policies. They collaborate with the program manager for quality improvement initiatives, patient safety, medical staff training and education.
2. Trauma Program Director/Program Manager (TPM)
Designate a licensed registered nurse experienced in the clinical care of adult and/or pediatric care of a trauma patient and provides administrative support, coordination, and collaboration with the trauma program medical director to ensure quality improvement, education, and training of the trauma team, ancillary staff and patients. They are responsible for compliance with State statute, regulation, and County policy along with the program Medical Director.
3. Trauma Team
Designate a multidisciplinary group of personnel to collectively render care for trauma patients. Team members are trained in initial resuscitation and management of the trauma patient.
4. Registrar(s)
Dedicated data expert(s), with current required certifications to collect, maintain and report data on trauma patients to support quality improvement, education, injury prevention and research initiatives of the trauma center. Must have an adequate number of trained and certified staff to meet volume of entries and ensure timely data



submission. It is recommended to have one full-time equivalent for each 400-600 patients abstracted into the registry.

5. Performance Improvement (PI) Coordinator(s)

Dedicated licensed registered nurse who is responsible for trauma process and outcome measures to identify areas for improvement and implement strategies to enhance clinical practice and improve patient safety. Must have at least 05. FTE dedicated to trauma performance improvement.

6. Injury Prevention Professional

Prioritizes injury prevention based on trends identified in trauma data and develops patient and community education, training and other outreach activities (L1, must be other than the TPM or PI personnel).

D. Trauma System Participation

1. Participate in the pre-hospital performance improvement process and offer expertise to the EMS Medical Director in the development of pre-hospital care protocols relevant to trauma patients.
2. TPMD, TPM, and other specified individuals shall participate as members of the Trauma Care System Quality Improvement Committee (TCSQIC) as per Policy # 421 and other related committees when requested.
3. Participate in the pre-hospital quality improvement process through selected review of regional audit filters for trauma related deaths, major complications, and transfers when requested by the EMS Agency.
4. The Trauma Program Manager will work with the EMS Agency Specialty Programs Nurse Coordinator to obtain the necessary chart materials for the medical review process and committee meetings.
5. For any trauma center receiving a Trauma patient as a result of ED Bypass or Trauma Bypass shall conduct a quality review on the patient and report any adverse findings to the EMS Agency.
6. Participate in local and regional disaster/emergency management meetings, coalitions, and exercises at the request of the EMS Agency.



E. Data Collection and Submission

1. The EMS Agency will be responsible for creating a *Santa Clara County Trauma Data Dictionary* to include the required local, state, and national elements and will be updated annually.
2. The trauma center shall:
 - a. Collect the minimum data elements identified in the *Santa Clara County Trauma Data Dictionary* directly into an audited registry approved by the EMS Agency.
 - b. Submit the data in an agreed-upon format to the EMS Agency on regularly assigned dates.
 - c. Ensure that data is validated and without error prior to submission.
3. Trauma data will be submitted to the California EMS Information System as required by State regulations.

F. Performance Standards

1. Written EMS policies and procedures shall be reviewed at a minimum of every two (2) years but may be updated sooner based upon identified quality improvement needs.
2. Have a process to review and revise facility trauma policies and protocols at least every three (3) years.
3. Have a trauma performance improvement and patient safety (PIPS) plan that outlines the process for event identification, review, loop closure, and committee review with a multidisciplinary team.
4. Have a risk-adjusted benchmarking program to identify opportunities for improvement in patient care, and data quality.
5. Additional performance measures as determined by the EMS Agency and/or the TCSQIC.



G. Education

1. Document public education and injury prevention outreach activities which align with trauma system goals and submit an annual summary of these activities to the EMS Agency by the end of each calendar year.
2. Ensure staff physicians, staff nurses, staff allied health personnel, and members of the trauma team receive continuing education as required by ACS-COT.
3. Document pre-hospital and hospital continuing education activities and provide an annual summary of these activities to the EMS Agency by the end of each calendar year.
4. Provide trauma injury care education and guidance to non-trauma center(s) within the county.
5. Support a trauma research program (L1, PTC1 only).

VII. Reporting Requirements

1. Notify the EMS Agency Duty Chief as soon as possible in the case of an unplanned event causing trauma services to be unavailable as well as when the issue is resolved.
2. Notify the EMS Agency through the submission of a System Variance Report at least 24 hours prior to any planned event that will result in the need for trauma diversion.
3. Notify the EMS Agency in writing of any failure to meet these EMS Trauma Center Standards within 48 hours.
4. Changes to key trauma center personnel shall be reported to the EMS Agency within 10 (ten) business days to include:
 - a. Trauma Medical Director.
 - b. Trauma Program Manager.



VIII. References

- A. American College of Surgeons. Committee on Trauma. Resources for optimal care of the injured patient. American College of Surgeons, Committee on Trauma; 2022
- B. American College of Surgeons. Committee on Trauma. Resources for optimal care of the injured patient, change log. American College of Surgeons, Committee on Trauma; 2023